FEDERATED LEARNING OVER WEATHER STATION NETWORKS: PERSONALIZED TEMPERATURE FORECASTING FOR ENERGY DEMAND ESTIMATION

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ABSTRACT

Hint (remove before submission): *The abstract provides a concise summary of the project, including the FL application, empirical graph modelling, variation minimization approach, and the FL algorithms used. Do not include detailed numerical results. Instead, focus on problem, method, and evaluation setup.*

This report presents a federated learning (FL) system for next-day mean temperature forecasting across a network of nine Finnish Meteorological Institute (FMI) weather stations spanning coastal, inland, and sub-arctic climate regimes. Each station trains a personalized linear regression model on its local daily measurements, collaborating with geographically nearby stations via a weighted FL network without sharing raw data. Temperature predictions are translated into heating degree days (HDD) following the Finnish Meteorological Institute standard, providing a proxy for regional residential heating energy demand.

We formulate the problem as Generalized Total Variation Minimization (GTVMin) and solve it using synchronous federated gradient descent. Two FL systems are compared: one using a purely distance-based collaboration graph and one using a temperature-correlation-weighted graph, evaluated on a held-out test set using mean squared error.

***Index Terms*—** Federated learning, networks, personalized machine learning, trustworthy artificial intelligence

Hint (remove before submission): *Your report must adhere to the following formatting and length requirements:*

- *Use the LaTeX settings as defined in this file without changing margins or font sizes.*
- *You must not modify the number or titles of sections in this template.*
- *The report must not exceed 5 pages in total. The 5^{th} page must contain only the references.*

Hint (remove before submission): *You are encouraged to use terminology and notation consistent with the Aalto Dictionary of Machine Learning [**?**]. The dictionary repository also contains 300+ reusable TikZ figures (FL networks, graphs, loss landscapes, etc.) that you can copy into your report.*

1. INTRODUCTION

Hint (remove before submission): *Describe the application scenario, motivation, and learning objective of your federated learning (FL) system. Your introduction should include:*

- *a clear **motivation** for why FL is suitable for your application,*
- *sufficient **background** on the problem domain and related work,*
- *a concise **outline** of the report structure (one sentence per section).*

Accurate short-term temperature forecasting is a critical input for regional energy demand planning. Utilities and grid operators rely on next-day temperature predictions to estimate residential heating requirements and balance load across regions.

In Finland, heating demand varies substantially across the country: coastal areas benefit from the moderating effect of the Baltic Sea and its bays, while inland and northern regions experience lower and more volatile temperatures. A single global forecasting model trained on all stations simultaneously would fail to capture these local characteristics. Conversely, purely local models trained on a single station's historical record may be unreliable, particularly for stations with sparse or noisy measurements. Federated Learning (FL) offers a solution to this [1]. In an FL system, each device - here, each weather station - trains its own personalized model on local data while collaborating with neighboring devices by exchanging model parameters rather than raw measurements. This makes FL both privacy-friendly and decentralized: no raw meteorological data needs to leave each station. FL has been proposed for smart grid applications, where heterogeneous local conditions and data privacy requirements make centralized approaches impractical [1]. This project implements an FL system over a network of nine Finnish Meteorological Institute (FMI) weather stations to predict next-day mean temperature at each station. Predictions are translated into heating degree days (HDD) following the Finnish Meteorological Institute standard, providing a proxy for regional residential heating energy demand. The FL system is formulated as Generalized Total Variation Minimization (GTVMin) [1], which balances local model fit against inter-station agreement, with collaboration strength proportional to geographic proximity. The remainder of this report is structured as follows. Section 2 formulates the FL problem, including network design, feature and label definitions, local models, and loss functions. Section 3 describes the GTVMin-based methodology and the federated gradient descent algorithm. Section 4 presents numerical experiments comparing two FL system designs. Section 5 concludes with a brief discussion of results and directions for future work.

2. PROBLEM FORMULATION

Hint (remove before submission): *All projects use the same raw data source and node definition:*

- ***Raw data:*** *Finnish Meteorological Institute (FMI) Open Data (`https://en.ilmatieteenlaitos.fi/open-data`).*
- ***Nodes (fixed):*** *each node in the FL network is one FMI weather station.*

A Python script for fetching FMI data is available here:

*Everything else is **your design choice**. You must clearly describe and justify:*

- *which stations you selected and why,*
- *how edges between stations are defined and how edge weights are set (e.g., geographic proximity, data similarity); see [**?**, Ch. 7] for data-driven methods,*
- *how local data points are derived from raw FMI measurements (choice of features and labels),*
- *what model is trained locally at each station,*
- *how the local loss functions are defined.*

Raw data. We use open meteorological measurements from the Finnish Meteorological Institute (FMI) Open Data API [5]. We select daily observations from January 2022 to December 2024, covering three full years. The measured variables are: daily mean temperature (`TDAY`), daily minimum temperature (`TMIN`), daily maximum temperature (`TMAX`), and daily mean wind speed (`WS_10MIN`).

Station selection. We select nine FMI stations representing three climatically distinct regimes across Finland:

- **Coastal:** Helsinki Kaisaniemi, Turku Artukainen, Oulu Vihreäsaari
- **Inland:** Tampere Härmälä, Jyväskylä, Kuopio Maaninka
- **Northern:** Rovaniemi Apukka, Sodankylä, Inari Saariselkä

This selection spans approximately 60°N to 68°N latitude, capturing maritime south-west, continental central, and sub-arctic climate regimes. The nine-station network is compact enough for full analysis within project scope while large enough to produce meaningful variation in local model parameters. Stations were selected to ensure geographic spread and avoid clustering, so that the FL network captures genuine climate heterogeneity rather than redundant nearby measurements.

Local data points. Each node i holds a local dataset $\mathcal{D}^{(i)} = \{(\mathbf{x}^{(i,t)}, y^{(i,t)})\}_{t=1}^{m_i}$, where the feature vector $\mathbf{x}^{(i,t)} \in \mathbb{R}^4$ for day t is:

$$\mathbf{x}^{(i,t)} = \left[T_{\text{mean}}^{(t)},\ T_{\text{min}}^{(t)},\ T_{\text{max}}^{(t)},\ \text{WS}^{(t)}\right]^\top \tag{1}$$

and the label $y^{(i,t)} = T_{\text{mean}}^{(t+1)}$ is the next-day mean temperature. Using today's mean, minimum, maximum temperature, and wind speed as features is motivated by their direct physical relationship to next-day temperature: the daily temperature range captures atmospheric stability, while wind speed influences advective heat transport. Predicting temperature directly is preferred over predicting HDD directly because temperature is a clean, continuously measured FMI variable with well-understood statistical properties, making it more suitable as a regression target.

Heating degree days are recovered from predictions as a post-processing step, following the Finnish Meteorological Institute standard [6]:

$$\text{HDD} = \begin{cases} \max(17 - \hat{y},\, 0) & \text{if } \hat{y} < 12°\text{C} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

where $\hat{y}$ is the predicted next-day mean temperature. This standard uses an indoor baseline of 17°C and a heating activation threshold of 12°C, reflecting that Finnish buildings do not require active heating above this threshold due to internal heat gains from occupants and appliances [6]. This 1-day-ahead forecasting horizon is motivated by operational energy planning requirements.

FL network. Each node $i \in \mathcal{V}$ corresponds to one FMI station. Edges $\{i, i'\} \in \mathcal{E}$ are defined between all pairs of stations within a geographic distance threshold of $d_{\max} = 300\,\text{km}$, computed using the Haversine formula from station coordinates. Edge weights are set as:

$$A_{i,i'} = \exp\left(-\frac{d(i,i')^2}{2\sigma^2}\right), \quad \sigma = 150\,\text{km} \tag{3}$$

where $d(i, i')$ is the great-circle distance between stations i and i'. This Gaussian kernel assigns higher collaboration weight to nearby stations and decays smoothly with distance, reflecting the spatial autocorrelation of temperature fields. The bandwidth $\sigma = 150\,\text{km}$ is set to half the distance threshold, ensuring that stations at the boundary of the neighbourhood receive a weight of $\exp(-2) \approx 0.14$, providing weak but non-zero collaboration. Stations beyond 300 km receive no edge. The values $\sigma = 150\,\text{km}$ and $d_{\max} = 300\,\text{km}$ are fixed design choices: $d_{\max}$ is set to ensure each station has at least two neighbors given the selected station layout, and σ is set to half the distance threshold so that boundary neighbors receive a non-negligible weight of $\exp(-2) \approx 0.14$. This distance-based graph defines **System A** and serves as the baseline for comparison in Section 4.

Local models. Each node i trains a linear regression model with parameter vector $\mathbf{w}^{(i)} \in \mathbb{R}^4$ (plus bias term), producing predictions $\hat{y}^{(i)} = \mathbf{x}^\top \mathbf{w}^{(i)} + b^{(i)}$. Linear models are chosen for two reasons: first, they are interpretable - the learned weights directly reflect how each station weighs temperature range and wind speed in its local forecast; second, daily mean temperature is well-approximated as a linear combination of same-day meteorological measurements [2].

Local loss functions. Each node minimises the mean squared error (MSE) on its local dataset:

$$L^{(i)}(\mathbf{w}^{(i)}) = \frac{1}{m_i}\left\|\mathbf{y}^{(i)} - \mathbf{X}^{(i)}\mathbf{w}^{(i)}\right\|_2^2 \tag{4}$$

MSE is chosen over mean absolute error (MAE) for two reasons. First, MSE penalises large prediction errors more heavily, which is desirable for energy planning where extreme cold events carry disproportionate consequences for heating demand. Second, MSE yields a smooth and strongly convex local loss function, which is a required property for the convergence guarantees of the federated gradient descent algorithm described in Section 3 [1].

3. METHODOLOGY

Hint (remove before submission): *The project requires you to apply GTVMin-based methods to the FL application modelled in Section 2. In this section you need to clearly state and explain:*

- *Your choice of variation measure, e.g., GTV penalty for parametric models.*
- *Your choice of FL algorithm (i.e., optimization method for solving GTVMin) and its synchronous message passing implementation.*

You are not expected to propose novel FL algorithms; correct implementation and analysis of existing GTVMin-based methods is sufficient.

Variation measure. We formulate the FL problem as Generalized Total Variation Minimization (GTVMin) [1], which balances local model fit at each station against agreement between neighboring stations. The variation between neighboring nodes i and i' is measured using the squared Euclidean distance between their local model parameters:

$$d\left(\mathbf{w}^{(i)}, \mathbf{w}^{(i')}\right) = \left\|\mathbf{w}^{(i)} - \mathbf{w}^{(i')}\right\|_2^2 \tag{5}$$

This choice is motivated by two properties. First, it is a natural measure of disagreement for real-valued parameter vectors. Second, combined with the MSE local loss functions defined in Section 2, it renders the full GTVMin objective smooth and strongly convex, which is a sufficient condition for the convergence of the federated gradient descent algorithm described below [1].

The full GTVMin objective over all $n = 9$ nodes is:

$$\min_{\mathbf{w}^{(1)},\ldots,\mathbf{w}^{(n)}} \underbrace{\sum_{i=1}^{n} L^{(i)}\Big(\mathbf{w}^{(i)}\Big)}_{\text{local fit}} + \alpha \underbrace{\sum_{\{i,i'\}\in\mathcal{E}} A_{i,i'} \Big\|\mathbf{w}^{(i)} - \mathbf{w}^{(i')}\Big\|_2^2}_{\text{GTV penalty}} \tag{6}$$

where $\alpha > 0$ is the regularization parameter controlling the trade-off between local fit and inter-station agreement. When $\alpha = 0$, each station trains independently with no collaboration; as $\alpha \to \infty$, all stations are forced toward a single shared global model. An intermediate value of α allows stations to maintain personalized models while benefiting from collaboration with neighbors, which is the FL design principle motivating this work [1]. The value of α is treated as a hyperparameter and selected by minimizing validation loss independently for each FL system.

FL algorithm. We solve GTVMin via federated gradient descent, derived as a fixed-point iteration of the GTVMin optimality conditions [1]. The gradient of the GTVMin objective with respect to node i's parameters decomposes into a local gradient and a coupling term that depends on neighbors:

$$\mathbf{g}^{(i)}(\mathbf{w}^{(t)}) = \nabla L^{(i)}\Big(\mathbf{w}^{(i,t)}\Big) + 2\alpha \sum_{i'\in\mathcal{N}(i)} A_{i,i'}\Big(\mathbf{w}^{(i,t)} - \mathbf{w}^{(i',t)}\Big) \tag{7}$$

At each iteration t, every node i updates its local parameters as:

$$\mathbf{w}^{(i,t+1)} = \mathbf{w}^{(i,t)} - \eta\, \mathbf{g}^{(i)}(\mathbf{w}^{(t)}) \tag{8}$$

where $\eta > 0$ is the learning rate. The update has a clear interpretation: each station takes a gradient step to improve its local temperature forecast, then adjusts its parameters toward those of its neighbors in proportion to the edge weight $A_{i,i'}$. Stations with strong edges (nearby, similar climate) exert more pull; distant stations exert none.

The algorithm is implemented as synchronous message passing in three steps per iteration [1]:

1. **Share:** every node i broadcasts its current parameters $\mathbf{w}^{(i,t)}$ to all neighbors $i' \in \mathcal{N}(i)$.
2. **Update:** every node i computes $\mathbf{g}^{(i)}(\mathbf{w}^{(t)})$ and performs the gradient step to obtain $\mathbf{w}^{(i,t+1)}$.
3. **Repeat** until a convergence criterion is met or a maximum number of iterations T is reached.

No raw meteorological data is exchanged at any step - only model parameter vectors $\mathbf{w}^{(i)} \in \mathbb{R}^4$, making the system privacy-friendly by design. All nodes are initialized at $\mathbf{w}^{(i,0)} = \mathbf{0}$ for $i = 1, \ldots, n$. The learning rate η, maximum iterations T, and convergence tolerance ϵ are treated as hyperparameters selected on the validation set.

Two FL systems compared. To evaluate the impact of graph construction on forecasting performance, we compare two FL systems that differ in a structural design choice - the definition of edge weights - while keeping the local models, loss functions, and algorithm identical:

- **System A - Distance graph** (baseline): edge weights defined purely by geographic proximity using the Gaussian kernel described in Section 2. This graph encodes the assumption that spatially close stations share similar temperature dynamics.

- **System B - Correlation graph**: edge weights are rescaled by the Pearson correlation of daily mean temperatures between station pairs, computed on the training set:

$$A_{i,i'}^{(B)} = \max\Big(0,\ \rho\Big(T^{(i)}, T^{(i')}\Big)\Big) \cdot \exp\left(-\frac{d(i,i')^2}{2\sigma^2}\right) \tag{9}$$

  Pairs with negative or zero correlation receive no edge. This graph encodes the assumption that stations with empirically similar temperature dynamics should collaborate more strongly, even if the distance-only weight does not reflect this.

System B is motivated by the observation that geographic proximity does not always imply climate similarity - for example, a coastal station and a nearby inland station may have systematically different temperature profiles due to sea influence. By comparing Systems A and B, we isolate the effect of graph construction on FL performance, which is a core design question in GTVMin-based FL [1].

4. NUMERICAL EXPERIMENTS

Hint (remove before submission): *Present quantitative results (training, validation, and test losses), comparisons, and figures. At a minimum, you must evaluate and compare* ***two distinct FL systems****. These systems must differ in a* structural design choice*, such as:*

- *different graph constructions or edge weights,*
- *different local model classes or model capacities,*
- *different local loss functions.*

Varying only hyper-parameters (e.g., regularization strength or step size) does ***not*** *count as a distinct FL system. Clearly explain evaluation metrics (e.g., which loss function has been used for training, validation, and final evaluation on the test set) and experimental settings (e.g., how data is split into train/validation/test sets and how hyper-parameters are selected). Briefly explain how the nature of the data points was taken into account when choosing the splitting strategy (e.g., chronological splitting for time-series data).*

Hint (remove before submission): Figures and tables. *Figures should be designed to communicate quantitative results clearly and efficiently. Avoid decorative elements that do not convey information (e.g., excessive colours, 3D effects, or chart junk). Axes, legends, and captions must be self-contained and precise. Whenever possible, embed numerical values directly into figures or tables rather than relying on verbose textual explanations. For general principles of effective quantitative graphics, see [?].*

Hint (remove before submission): No Python source code is required for submission. *Instead, the report itself must provide sufficient detail so that the FL algorithm can be* ***reimplemented using a large language model (LLM)*** *such as Claude Code. In particular, a reader should be able to paste the relevant sections into an LLM and obtain working code. Ensure you specify:*

- ***Data:*** *exact FMI station IDs (or names), time range, measured variables, and how features/labels are derived from raw measurements.*
- ***FL network:*** *which stations are connected and how edge weights are computed (with enough detail to recompute them).*
- ***Algorithm:*** *the update equation, learning rate, number of iterations, regularization parameter, and initialization.*

5. CONCLUSION

Hint (remove before submission): *Interpret the numerical results from Sec. 4. In particular,*

- *Discuss whether the obtained results solve the problem satisfactorily.*
- *Identify limitations of the studied FL methods and suggest potential improvements (e.g., collecting more data points or using larger local models at some nodes).*

You can find some guidance on how to diagnose ML models in [?, Sec. 6.6].

6. REFERENCES

[1] A. Jung, *Federated Learning: From Theory to Practice*, Springer, 2026. Preprint: `https://arxiv.org/abs/2505.19183`.

[2] A. Jung, *Machine Learning: The Basics*, Springer, 2022. Available via Aalto Library. Preprint: `https://github.com/alexjungaalto/MachineLearningTheBasics/`.

[3] A. Jung (Editor), *Aalto Dictionary of Machine Learning*, `https://aaltodictionaryofml.github.io/`, 2026.

[4] E. R. Tufte, *The Visual Display of Quantitative Information*, 2nd edition, Graphics Press, 2001.

[5] Finnish Meteorological Institute, “FMI Open Data,” `https://en.ilmatieteenlaitos.fi/open-data`, accessed May 2026.

[6] Finnish Meteorological Institute, “Heating Degree Days,” `https://en.ilmatieteenlaitos.fi/heating-degree-days`, accessed May 2026.